Guidance and Control of Unmanned Autonomous Vehicles

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Quaternions

- Complex numbers $\mathbb{C} = \{a + bi |\ a,\ b \in \mathbb{R},\ i^2 = -1\}$ form a plane.
- Their operations are related to two-dimensional geometry.
- Any complex number has a length, given by the Pythagorean formula

$$|a + bi| = \sqrt{a^2 + b^2}$$

- We can add and subtract in $\mathbb{C}$.

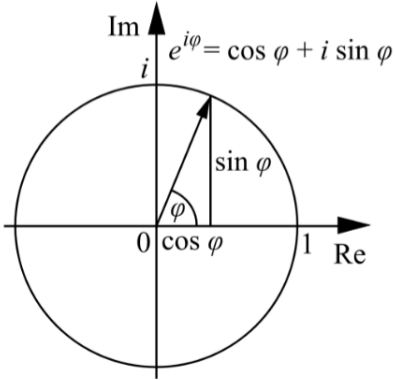
$$a + bi + c + di = (a + c) + (b + d)i$$

- We can also multiply in $\mathbb{C}$.

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

- What does this last formula mean? difficult to interpret
- Fortunately, there is a better way to multiply complex numbers.

Quaternions



We can use complex arithmetic (multiplication) to perform a geometric operation (rotation).

- Geometrically, this formula says $e^{i\phi}$ lies on the unit circle in $\mathbb{C}$.
- If we multiply $e^{i\phi}$ by a positive number r, we get a complex number of length r, $re^{i\phi}$.
- If we denote $a + bi = r_1 e^{i\theta_1}$ and $c + di = r_2 e^{i\theta_2}$ then

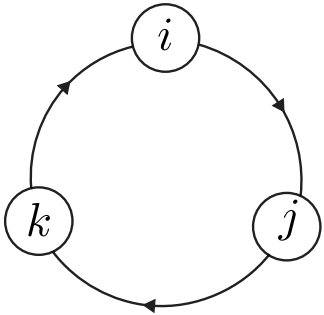
$$(a + bi)(c + di) = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

- To multiply two complex numbers, multiply their lengths and add their angles.
- In particular, if we multiply a given complex number z by $e^{i\phi}$ then it is rotated by ϕ degrees.

Quaternions: History

- The 19th century Irish mathematician and physicist William Rowan Hamilton was fascinated by the role of $\mathbb{C}$ in two-dimensional geometry.
- For years, he tried to invent an algebra of "triplets" to play the same role in three dimensions.
- On October 16th, 1843, while walking with his wife to a meeting of the Royal Society of Dublin, Hamilton discovered a 4-D division algebra, (called as quaternions).
- Although, a similar concept was developed by Gauss in 1819 (but unfortunately not published).

Quaternions



- Hamilton noticed that $\boxed{\boldsymbol{i}^2 = \boldsymbol{j}^2 = \boldsymbol{k}^2 = \boldsymbol{ijk} = -1}$
- Quaternions: $\mathbb{H} = \{a + b\boldsymbol{i} + c\boldsymbol{j} + d\boldsymbol{k} : a, b, c, d \in \mathbb{R}\}$

Quaternions

- Cyclic symmetry:

$$\boxed{\boldsymbol{i} \times \boldsymbol{j} = \boldsymbol{k} = -\boldsymbol{j} \times \boldsymbol{i}, \quad \boldsymbol{j} \times \boldsymbol{k} = \boldsymbol{i} = -\boldsymbol{k} \times \boldsymbol{j}, \quad \boldsymbol{k} \times \boldsymbol{i} = \boldsymbol{j} = -\boldsymbol{i} \times \boldsymbol{k}}$$

- The quaternion product is the same as the cross product of vectors.
- Quaternions don't commute.
- Vectors $\boldsymbol{i}, \boldsymbol{j}$ and $\boldsymbol{k}$ are recognized as unit vectors.
- Does it also follow all rules of the cross product for vectors? Not really
- Unlike the unit vectors, $\boxed{\boldsymbol{i} \times \boldsymbol{i} = \boldsymbol{j} \times \boldsymbol{j} = \boldsymbol{k} \times \boldsymbol{k} = 0}$, we have

$$\boxed{\boldsymbol{i} \times \boldsymbol{i} = -1, \quad \boldsymbol{j} \times \boldsymbol{j} = -1, \quad \boldsymbol{k} \times \boldsymbol{k} = -1, \quad \boldsymbol{i} \times \boldsymbol{j} \times \boldsymbol{k} = -1}$$

- **Quaternion algebra**: A mixture of algebra for vectors and complex numbers.

Quaternions

- A Hamilton quaternion can be considered as a scalar part and a vector part

$$\boxed{[Q] = \langle q_0, \boldsymbol{q} \rangle = q_0 + \boldsymbol{q} = q_0 + \underbrace{q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k}}_{\boldsymbol{q}}}$$

- $1, \boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ serves as basis for quaternion vector space.
- Quaternions span the space of real and imaginary numbers.
- Quaternion algebra includes scalar and vector algebra.
- **Addition, subtraction and multiplication**: Similar way as in vector algebra.
- **Addition**:

$$\begin{aligned}[Q] + [S] =& (q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k}) + (s_0 + s_1\boldsymbol{i} + s_2\boldsymbol{j} + s_3\boldsymbol{k}) \\ =& (q_0 + s_0) + (q_1 + s_1)\boldsymbol{i} + (q_2 + s_2)\boldsymbol{j} + (q_3 + s_3)\boldsymbol{k}\end{aligned}$$

Quaternions

- **Subtraction**: Addition of negative quaternion $-[Q] = (-1)[Q]$

$$\begin{aligned}[Q] - [S] =& (q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k}) - (s_0 + s_1\boldsymbol{i} + s_2\boldsymbol{j} + s_3\boldsymbol{k}) \\ =& (q_0 - s_0) + (q_1 - s_1)\boldsymbol{i} + (q_2 - s_2)\boldsymbol{j} + (q_3 - s_3)\boldsymbol{k}\end{aligned}$$

- **Multiplication**:

$$[Q][S] = (q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k})(s_0 + s_1\boldsymbol{i} + s_2\boldsymbol{j} + s_3\boldsymbol{k})$$

$$\begin{aligned}[Q][S] = \ & (q_0s_0 - q_1s_1 - q_2s_2 - q_3s_3) + (q_0s_1 + q_1s_0 + q_2s_3 - q_3s_2)\boldsymbol{i} \\ & + (q_0s_2 - q_1s_3 + q_2s_0 + q_3s_1)\boldsymbol{j} + (q_0s_3 + q_1s_2 - q_2s_1 + q_3s_0)\boldsymbol{k}\end{aligned}$$

- Using dot product of vectors $\boxed{\boldsymbol{q}.\boldsymbol{s} = q_1s_1 + q_2s_2 + q_3s_3}$, we have

$$\boxed{[Q][S] = \langle q_0, \boldsymbol{q}\rangle\langle s_0, \boldsymbol{s}\rangle = \langle q_0s_0 - \boldsymbol{q}.\boldsymbol{s}, q_0\boldsymbol{s} + s_0\boldsymbol{q} + \boldsymbol{q}\times\boldsymbol{s}\rangle}$$

Quaternions

- Product of two quaternions is still a quaternion, with a scalar part $(q_0 s_0 - \boldsymbol{q} \cdot \boldsymbol{s})$ and a vector part $(q_0 \boldsymbol{s} + s_0 \boldsymbol{q} + \boldsymbol{q} \times \boldsymbol{s})$.
- The set of quaternions is closed under multiplication and addition.

Example

Consider the two quaternions below and find their quaternion product.

$$[Q] = 3 + \boldsymbol{i} - 2\boldsymbol{j} + \boldsymbol{k}, \quad [S] = 2 - \boldsymbol{i} + 2\boldsymbol{j} + 3\boldsymbol{k}$$

$$\boldsymbol{q}.\boldsymbol{s} = -2, \quad \boldsymbol{q} \times \boldsymbol{s} = -8\boldsymbol{i} - 4\boldsymbol{j}$$

$$\langle q_0, \boldsymbol{q} \rangle \langle s_0, \boldsymbol{s} \rangle = \langle q_0 s_0 - \boldsymbol{q}.\boldsymbol{s}, q_0 \boldsymbol{s} + s_0 \boldsymbol{q} + \boldsymbol{q} \times \boldsymbol{s} \rangle$$

$$\langle q_0, \boldsymbol{q} \rangle \langle s_0, \boldsymbol{s} \rangle = 6 - (-2) + 3(-\boldsymbol{i} + 2\boldsymbol{j} + 3\boldsymbol{k}) + 2(\boldsymbol{i} - 2\boldsymbol{j} + \boldsymbol{k}) + (-8\boldsymbol{i} - 4\boldsymbol{j}) = 8 - 9\boldsymbol{i} - 2\boldsymbol{j} + 11\boldsymbol{k}$$

Quaternions

- **Scalar multiplication**:

$$\boxed{\lambda[Q] = \lambda q_0 + \lambda q_1 \boldsymbol{i} + \lambda q_2 \boldsymbol{j} + \lambda q_3 \boldsymbol{k}}$$

- **Conjugate**:

$$\boxed{[Q]^\star = q_0 - \boldsymbol{q} = q_0 - q_1 \boldsymbol{i} - q_2 \boldsymbol{j} - q_3 \boldsymbol{k}}$$

- We have the following relations

$$\begin{aligned}
([Q]^\star)^\star &= q_0 - (-\boldsymbol{q}) = [Q] \\
[Q] + [Q]^\star &= 2q_0 \\
[Q]^\star[Q] &= (q_0 + \boldsymbol{q})(q_0 - \boldsymbol{q}) = q_0 q_0 - (-\boldsymbol{q}).\boldsymbol{q} + q_0\boldsymbol{q} + (-\boldsymbol{q})q_0 + (-\boldsymbol{q} \times \boldsymbol{q}) \\
&= q_0^2 + \boldsymbol{q}.\boldsymbol{q} = q_0^2 + q_1^2 + q_2^2 + q_3^2 = [Q][Q]^\star
\end{aligned}$$

Quaternions

- **Norm or Length of quaternion**:

$$
\begin{aligned}
N([Q]) = [Q][Q]^\star = [Q]^\star[Q] &= (q_0 + \boldsymbol{q})(q_0 - \boldsymbol{q}) \\
&= q_0 q_0 - (-\boldsymbol{q}).\boldsymbol{q} + q_0\boldsymbol{q} + (-\boldsymbol{q})q_0 + (-\boldsymbol{q} \times \boldsymbol{q}) \\
&= q_0^2 + \boldsymbol{q}.\boldsymbol{q} = q_0^2 + q_1^2 + q_2^2 + q_3^2
\end{aligned}
$$

- For two quaternions $[Q]$ and $[S]$, $\boxed{([Q][S])^\star = [S]^\star[Q]^\star}$

▶ Proof:

$$
\begin{aligned}
([Q][S])^\star &= [(q_0 + \boldsymbol{q})\,(s_0 + \boldsymbol{s})]^\star = (q_0 s_0 - \boldsymbol{q}.\boldsymbol{s} + q_0\boldsymbol{s} + s_0\boldsymbol{q} + \boldsymbol{q} \times \boldsymbol{s})^\star \\
&= (q_0 s_0 - \boldsymbol{q}.\boldsymbol{s} - q_0\boldsymbol{s} - s_0\boldsymbol{q} - \boldsymbol{q} \times \boldsymbol{s})
\end{aligned}
$$

$$
\begin{aligned}
[S]^\star[Q]^\star &= (s_0 - \boldsymbol{s})\,(q_0 - \boldsymbol{q}) = q_0 s_0 - (-\boldsymbol{s}).(-\boldsymbol{q}) + q_0(-\boldsymbol{s}) + s_0(-\boldsymbol{q}) + (-\boldsymbol{s}) \times (-\boldsymbol{q}) \\
&= (q_0 s_0 - \boldsymbol{q}.\boldsymbol{s} - q_0\boldsymbol{s} - s_0\boldsymbol{q} - \boldsymbol{q} \times \boldsymbol{s}) = ([Q][S])^\star
\end{aligned}
$$

Quaternions

Identities:

- How to define zero and unit quaternions?
- A zero quaternion is quaternion with zero scalar and zero vector.
- A unit quaternion is defined as any quaternion whose norm is 1.

$$[0] = 0 + 0\boldsymbol{i} + 0\boldsymbol{j} + 0\boldsymbol{k}, \quad [1] = 1 + 0\boldsymbol{i} + 0\boldsymbol{j} + 0\boldsymbol{k}$$

- Unlike the direction cosine matrix, where six redundancies are present, the quaternion has only one.
- For unit quaternion, $q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$
- $\sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}$ can be used for a normalizing factor for each parameter.

Quaternions

- Norm of product of two quaternions is equal to product of their norms.

$$N([Q][S]) = N([Q])N([S])$$

$$N([Q][S]) = ([Q][S])([Q][S])^\star = [Q][S][S]^\star[Q]^\star = [Q]N([S])[Q]^\star = N([Q])N([S])$$

- Also, by using mathematical induction, one may write

$$\boxed{N([Q_1][Q_2]\ldots[Q_n]) = N([Q_1])N([Q_2])\ldots N([Q_n])}$$

- **Inverse of quaternion**: If $[Q] \neq 0$ then its inverse is defined by

$$[Q][Q]^{-1} = [Q]^{-1}[Q] = 1$$

- Using norm concept, $[Q]^{-1} = \dfrac{[Q]^\star}{N([Q])}$, $N([Q]) \neq 0$. Does it make sense?

$$[Q][Q]^{-1} = [Q]^{-1}[Q] = \frac{[Q][Q]^\star}{N([Q])} = 1$$

Quaternions

- Using the norm concept,

$$[Q][Q]^{-1} = [Q]^{-1}[Q] = \frac{[Q][Q]^{\star}}{N(Q)} = 1$$

- If $[Q]$ is unit quaternion then

$$[Q]^{-1} = \frac{[Q]^{\star}}{N(Q)} = [Q]^{\star}$$

- **Inverse and conjugate for the unit quaternions are the same.**
- How to identify if two quaternions are equal?
- **Equality of quaternions**: Two quaternions are equal if both their scalars as well as their vectors are equal.

$$[Q] = [S] \implies q_0 = s_0,\ q_1 = s_1,\ q_2 = s_2,\ q_3 = s_3$$

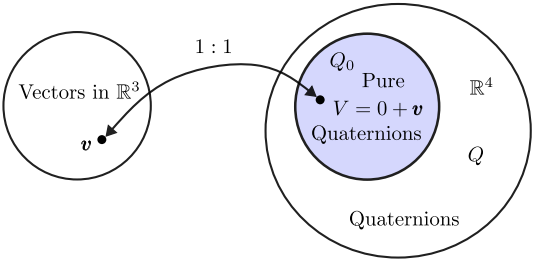
Quaternions

- Quaternions obey the associative and commutative laws of addition, and the associative and distributive laws of multiplication.
- For three quaternions, $[Q_1], [Q_2], [Q_3]$
  - ☐ Associative addition: $([Q_1] + [Q_2]) + [Q_3] = [Q_1] + ([Q_2] + [Q_3])$
  - ☐ Commutative addition: $[Q_1] + [Q_2] = [Q_2] + [Q_1]$
  - ☐ Associative multiplication: $([Q_1][Q_2])[Q_3] = [Q_1]([Q_2][Q_3])$
  - ☐ Distributive multiplication: $[Q_1]([Q_2] + [Q_3]) = [Q_1][Q_2] + [Q_1][Q_3]$
- Is the multiplication of quaternions commutative? No

$$[Q][S] \neq [S][Q]$$

Quaternion

- How can a quaternion $[Q] \in \mathbb{R}^4$ operate on a vector $\boldsymbol{v} \in \mathbb{R}^3$?
- A vector $\boldsymbol{v} \in \mathbb{R}^3$ treated as a quaternion with zero real part.
- Pure quaternions, $[V] = 0 + \boldsymbol{v}$



- Will a rotation operator defined in terms of quaternions have the same form as a matrix rotation operator?
- For a quaternion $[Q]$, can we write

$$\boldsymbol{w} = [Q]\boldsymbol{v}?$$

- **Requirement**: Product of a quaternion with a vector must not only always be defined, but the result must always be a vector.
- Product of a quaternion with a vector using pure quaternion.

Quaternion

- For given $[Q] = q_0 + \boldsymbol{q}$ and $\boldsymbol{v}$,

$$
\begin{aligned}
[Q]\boldsymbol{v} &= (q_0 + \boldsymbol{q})(0 + \boldsymbol{v}) \\
&= q_0.0 - \boldsymbol{q}.\boldsymbol{v} + 0.\boldsymbol{q} + q_0\boldsymbol{v} + \boldsymbol{q} \times \boldsymbol{v} \\
&= -\boldsymbol{q}.\boldsymbol{v} + q_0\boldsymbol{v} + \boldsymbol{q} \times \boldsymbol{v}
\end{aligned}
$$

- Not a vector unless $\boldsymbol{q}.\boldsymbol{v} = 0$
- Orthogonal $\boldsymbol{q}$ and $\boldsymbol{v}$
- $\boldsymbol{v}[Q]$ does not work as rotation operator

$$
\boldsymbol{v}[Q][R], [Q][R]\boldsymbol{v}, [R]\boldsymbol{v}[Q]
$$

$$
\boldsymbol{v}[R][Q], [R][Q]\boldsymbol{v}, [Q]\boldsymbol{v}[R]
$$

- Set of quaternions is closed under multiplication.
- $[Q][R],\ [R][Q]$ result in quaternion and thus only two possibilities $[R]\boldsymbol{v}[Q], [Q]\boldsymbol{v}[R]$
- Real part of product with $[Q] = q_0 + \boldsymbol{q}, [R] = r_0 + \boldsymbol{r}, [P] = 0 + \boldsymbol{p}$,

$$
\begin{aligned}
&- r_0(\boldsymbol{q}.\boldsymbol{p}) - q_0(\boldsymbol{p}.\boldsymbol{r}) - (\boldsymbol{q} \times \boldsymbol{p}).\boldsymbol{r} = \\
&- r_0(\boldsymbol{q}.\boldsymbol{p}) - q_0(\boldsymbol{p}.\boldsymbol{r}) - (\boldsymbol{q} \times \boldsymbol{r}).\boldsymbol{p}
\end{aligned}
$$

- Cyclic property of vector product

$$
\boxed{\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b} \cdot (\boldsymbol{c} \times \boldsymbol{a}) = \boldsymbol{c} \cdot (\boldsymbol{a} \times \boldsymbol{b})}
$$

Quaternion

- How to ensure that the real part is zero?
- With $r_0 = q_0$, real part

$$-q_0(\boldsymbol{q}+\boldsymbol{r}).\boldsymbol{p} + (\boldsymbol{q}\times\boldsymbol{r}).\boldsymbol{p}$$

- For real part to be zero, $\boldsymbol{r} = -\boldsymbol{q}$.

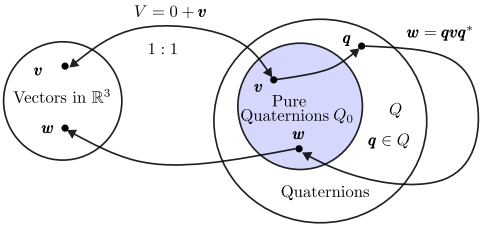
$$[R] = r_0 + \boldsymbol{r} = q_0 - \boldsymbol{q} = [Q]^* \implies \boxed{[Q] = [R]^*}$$

- Two possible triple products,

$$[Q]\boldsymbol{v}[Q]^*, \qquad [Q]^*\boldsymbol{v}[Q]$$

- Quaternion operator

$$[Q] = \cos\theta + \boldsymbol{u}\sin\theta,\ \boldsymbol{u} = \frac{\boldsymbol{q}}{\|\boldsymbol{q}\|} = \frac{\boldsymbol{q}}{\sin\theta}$$



Quaternions

- How quaternion $[Q] \in \mathbb{R}^4$ operates on a vector $\boldsymbol{v} \in \mathbb{R}^3$?
- Define quaternion operator with unit quaternion $[Q]$ as

$$L_Q(\boldsymbol{v}) = [Q]\boldsymbol{v}[Q]^\star = (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{v} + 2(\boldsymbol{q}.\boldsymbol{v})\boldsymbol{q} + 2q_0(\boldsymbol{q} \times \boldsymbol{v})$$

$$\begin{aligned}
L_Q(\boldsymbol{v}) =& [Q]\boldsymbol{v}[Q]^\star = (q_0 + \boldsymbol{q})(0 + \boldsymbol{v})(q_0 - \boldsymbol{q}) \\
=& (q_0 + \boldsymbol{q})(\boldsymbol{v}.\boldsymbol{q} + \{q_0\boldsymbol{v} - \boldsymbol{v} \times \boldsymbol{q}\}) \\
=& q_0(\boldsymbol{v}.\boldsymbol{q}) - \boldsymbol{q}.\{q_0\boldsymbol{v} - \boldsymbol{v} \times \boldsymbol{q}\} + (\boldsymbol{v}.\boldsymbol{q})\boldsymbol{q} + q_0\{q_0\boldsymbol{v} - \boldsymbol{v} \times \boldsymbol{q}\} + \boldsymbol{q} \times \{q_0\boldsymbol{v} - \boldsymbol{v} \times \boldsymbol{q}\} \\
=& \boldsymbol{q}.\{\boldsymbol{v} \times \boldsymbol{q}\} + (\boldsymbol{v}.\boldsymbol{q})\boldsymbol{q} + q_0\{q_0\boldsymbol{v} - \boldsymbol{v} \times \boldsymbol{q}\} + \boldsymbol{q} \times \{q_0\boldsymbol{v} - \boldsymbol{v} \times \boldsymbol{q}\} \\
=& (\boldsymbol{v}.\boldsymbol{q})\boldsymbol{q} + q_0^2\boldsymbol{v} + 2q_0\{\boldsymbol{q} \times \boldsymbol{v}\} + \boldsymbol{q} \times \{\boldsymbol{q} \times \boldsymbol{v}\} \\
=& (\boldsymbol{v}.\boldsymbol{q})\boldsymbol{q} + q_0^2\boldsymbol{v} + 2q_0\{\boldsymbol{q} \times \boldsymbol{v}\} + \boldsymbol{q}(\boldsymbol{q}.\boldsymbol{v}) - \boldsymbol{v}(\boldsymbol{q}.\boldsymbol{q}) \\
=& (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{v} + 2(\boldsymbol{q}.\boldsymbol{v})\boldsymbol{q} + 2q_0\{\boldsymbol{q} \times \boldsymbol{v}\}
\end{aligned}$$

Quaternions

Operation of Unit Quaternion on Vector

This operator L_Q does not change the length of the vector $\boldsymbol{v}$.

$$\|L_Q(\boldsymbol{v})\| = \|[Q]\boldsymbol{v}[Q]^\star\| = |[Q]|\|\boldsymbol{v}\||[Q]^\star| = \|\boldsymbol{v}\|$$

The direction of $\boldsymbol{v}$, if along $\boldsymbol{q}$ (say $\boldsymbol{v} = k\boldsymbol{q}$), is left unchanged by the operator L_Q.

$$[Q]\boldsymbol{v}[Q]^\star = [Q](k\boldsymbol{q})[Q]^\star = (q_0^2 - |\boldsymbol{q}|^2)(k\boldsymbol{q}) + 2(\boldsymbol{q}.(k\boldsymbol{q}))\boldsymbol{q} + 2q_0(\boldsymbol{q} \times (k\boldsymbol{q})) = k(q_0^2 + \|\boldsymbol{q}\|^2)\boldsymbol{q} = k\boldsymbol{q}$$

- Any vector along $\boldsymbol{q}$ is thus not changed under the operator L_Q.
- This makes us guess that the operator L_Q acts like a rotation about $\boldsymbol{q}$.
- Operator L_Q is linear over $\mathbb{R}^3$. For any two vectors $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^3$ and any $a_1, a_2 \in \mathbb{R}$

$$L_Q(a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2) = a_1 L_Q(\boldsymbol{v}_1) + a_2 L_Q(\boldsymbol{v}_2).$$

Quaternions

Rotation of Vector using Quaternion

For any unit quaternion $[Q] = q_0 + \boldsymbol{q} = \cos\frac{\theta}{2} + \hat{q}\sin\frac{\theta}{2}$ and for any vector $\boldsymbol{v} \in \mathbb{R}^3$, the action of the operator $L_Q(\boldsymbol{v}) = [Q]\boldsymbol{v}[Q]^\star$ on $\boldsymbol{v}$ is equivalent to a rotation of the vector through an angle θ, about $\hat{q}$ as the axis of rotation.

- A vector $\boldsymbol{v} \in \mathbb{R}^3$, we decompose it as $\boldsymbol{v} = \boldsymbol{a} + \boldsymbol{n}$, where $\boldsymbol{a}$ is the component along the vector $\boldsymbol{q}$ and $\boldsymbol{n}$ is the component normal to $\boldsymbol{q}$.
- Under the operator L_Q, $\boldsymbol{a}$ is invariant, while $\boldsymbol{n}$ is rotated about $\boldsymbol{q}$ through an angle θ.
- Since the operator is linear, the image $[Q]\boldsymbol{v}[Q]^\star$ is indeed interpreted as a rotation of $\boldsymbol{v}$ about $\boldsymbol{q}$ through an angle θ.

Quaternions

- The operator L_q on vectors $\boldsymbol{n}$

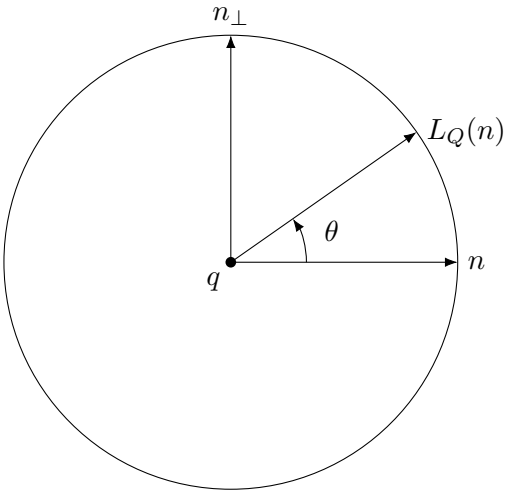
$$\begin{aligned} L_Q(\boldsymbol{n}) =& (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2(\boldsymbol{q}.\boldsymbol{n})\boldsymbol{q} + 2q_0(\boldsymbol{q} \times \boldsymbol{n}) \\ =& (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2q_0(\boldsymbol{q} \times \boldsymbol{n}) \\ =& (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2q_0\|\boldsymbol{q}\|(\hat{\boldsymbol{q}} \times \boldsymbol{n}) \end{aligned}$$

- Denoting $\boxed{\boldsymbol{n}_\perp = \hat{\boldsymbol{q}} \times \boldsymbol{n}}$, one can write

$$\begin{aligned} L_Q(\boldsymbol{n}) &= (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2q_0\|\boldsymbol{q}\|\boldsymbol{n}_\perp = \left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right)\boldsymbol{n} + 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\boldsymbol{n}_\perp \\ &= \cos\theta\boldsymbol{n} + \sin\theta\boldsymbol{n}_\perp \end{aligned}$$

- Resulting vector is a rotation of $\boldsymbol{n}$ through an angle θ in the plane defined by $\boldsymbol{n}$ and $\boldsymbol{n}_\perp$.

Quaternions



- $L_{-[Q]} = [-Q]\boldsymbol{v}[-Q]^\star = [Q]\boldsymbol{v}[Q]^\star$
- How? What does it mean?
- Negative quaternion $-[Q]$

$$-[Q] = \cos\frac{2\pi+\theta}{2} + \hat{q}\sin\frac{2\pi+\theta}{2}$$

- It represents the rotation about the same axis through the angle $2\pi+\theta$, essentially the same rotation.

Quaternions

Rotation of Coordinate Frame using Quaternion

For any unit quaternion $[Q] = q_0 + \boldsymbol{q} = \cos(\theta/2) + \hat{q}\sin(\theta/2)$ and for any vector $\boldsymbol{v} \in \mathbb{R}^3$, the action of the operator $L_{Q^\star}(\boldsymbol{v}) = [Q]^\star \boldsymbol{v}[Q]^{\star\star} = [Q]^\star \boldsymbol{v}[Q]$ is a rotation of the coordinate frame about the axis $\hat{q}$ through an angle θ while $\boldsymbol{v}$ is not rotated.

- Rotation of $\boldsymbol{v}$ under the operator L_Q can also be interpreted from the perspective of an observer attached to the vector $\boldsymbol{v}$.
- What he sees happening is that the coordinate frame rotates through the angle $-\theta$ about the same axis defined by the quaternion.
- $L_{Q^\star}$ rotates vector $\boldsymbol{v}$ w.r.t. coordinate frame through an angle $-\theta$ about $\boldsymbol{q}$.
- $L_Q(\boldsymbol{v}) = [Q]v[Q]^\star$: A point or vector rotation with respect to the (fixed) coordinate frame.
- $L_{Q^\star}(\boldsymbol{v}) = [Q]^\star \boldsymbol{v}[Q]$: A coordinate frame rotation with respect to the (fixed) space

Quaternions

- Quaternion operator

$$
\begin{aligned}
L_Q(\boldsymbol{v}) =& [Q]\boldsymbol{v}[Q]^\star = (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{v} + 2(\boldsymbol{q}\cdot\boldsymbol{v})\boldsymbol{q} + 2q_0(\boldsymbol{q}\times\boldsymbol{v}) \\
=& \left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right)\boldsymbol{v} + 2\left(\hat{\boldsymbol{q}}\sin\frac{\theta}{2}.\boldsymbol{v}\right)\hat{\boldsymbol{q}}\sin\frac{\theta}{2} + 2\cos\frac{\theta}{2}\left(\hat{\boldsymbol{q}}\sin\frac{\theta}{2}\times\boldsymbol{v}\right) \\
=& \cos\theta\boldsymbol{v} + (1-\cos\theta)\left(\hat{\boldsymbol{q}}\cdot\boldsymbol{v}\right)\hat{\boldsymbol{q}} + \sin\theta\left(\hat{\boldsymbol{q}}\times\boldsymbol{v}\right)
\end{aligned}
$$

- Quaternion operator in a matrix form

$$
L_Q(\boldsymbol{v}) = \underbrace{\left[(q_0^2 - \|\boldsymbol{q}\|^2)I_{3\times3} + 2\boldsymbol{q}\boldsymbol{q}^T + 2q_0\boldsymbol{q}^\times\right]}_{\text{Rotation Matrix}}\boldsymbol{v}
$$

where the matrix representing the cross-product is given by

$$
\boldsymbol{q}^\times = \begin{bmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{bmatrix}
$$

Quaternions

Rotation of Vector using Quaternion

Consider a rotation of a vector about an axis defined by (1, 1, 1) through an angle of $2\pi/3$. Obtain the quaternion to perform this rotation. Compute the effect of rotation on the basis vector $\boldsymbol{i} = (1, 0, 0)$.

- Define unit vector $\hat{q} = \dfrac{1}{\sqrt{3}}(1, 1, 1)$.
- Quaternion

$$[Q] = \cos\frac{\theta}{2} + \hat{q}\sin\frac{\theta}{2} = \frac{1}{2} + \frac{1}{2}\boldsymbol{i} + \frac{1}{2}\boldsymbol{j} + \frac{1}{2}\boldsymbol{k}$$

- Actual vector $\boldsymbol{v} = \boldsymbol{i} = (1, 0, 0)$.

Quaternions

- By using quaternion operator on $\boldsymbol{v} = (1, 0, 0)$, we get

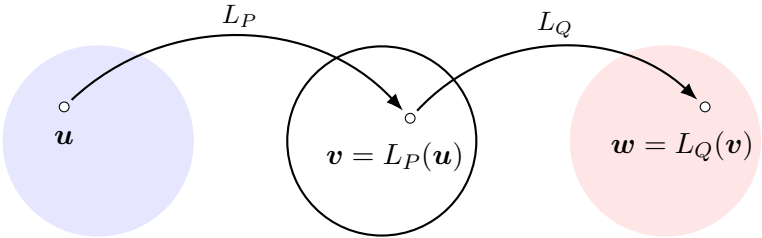
$$
\begin{aligned}
[w] &= \cos\theta \boldsymbol{v} + (1 - \cos\theta)\,(\hat{q}.\boldsymbol{v})\,\hat{q} + \sin\theta\,(\hat{q} \times \boldsymbol{v}) \\
&= -\frac{1}{2}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \left(1 + \frac{1}{2}\right)\frac{1}{\sqrt{3}}\frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{\sqrt{3}}{2}\frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} -1/2 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1/2 \\ -1/2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\
&= \boldsymbol{j}
\end{aligned}
$$

- How can we obtain an equivalent quaternion for a sequence of rotations?

Quaternions: Sequence of Quaternion Operators

- Let $[P]$ and $[Q]$ be two unit quaternions.

$$L_P(\boldsymbol{u}) = \boldsymbol{v}, \quad L_Q(\boldsymbol{v}) = \boldsymbol{w}$$



$$\boldsymbol{w} = L_Q(\boldsymbol{v}) = [Q]\boldsymbol{v}[Q]^\star = [Q][P]\boldsymbol{u}[P]^\star[Q]^\star = [QP]\boldsymbol{u}[QP]^\star = L_{QP}(\boldsymbol{u})$$

- L_{QP} is a unit quaternion rotation operator, with the axis and angle of the composite rotation given by the product $[QP]$.

Quaternions

- Consider quaternion operators $L_{P^\star}(\boldsymbol{u}) = [P]^\star \boldsymbol{u}[P]$ and $L_{Q^\star}(\boldsymbol{v}) = [Q]^\star \boldsymbol{v}[Q]$.
- These operators define rotations of the coordinate system defined by corresponding quaternions.

$$\boldsymbol{w} = L_{Q^\star}(\boldsymbol{v}) = [Q]^\star \boldsymbol{v}[Q] = [Q]^\star [P]^\star \boldsymbol{u}[P][Q] = [PQ]^\star \boldsymbol{u}[PQ] = L_{(PQ)^\star}(\boldsymbol{u})$$

- Quaternion product $([P][Q])^\star$ defines operator which represents a sequence of operators $L_{P^\star}$ followed by $L_{Q^\star}$.
- $L_{(PQ)^\star}$ is also a unit quaternion rotation operator, with the axis and angle of the composite rotation given by the product $[PQ]$.

Consider a rotation of the coordinate frame about the z-axis through an angle α, followed by a rotation about the new y-axis through an angle β. By using the quaternion method, find out the axis and angle of the composite rotation.

Quaternions

- The first rotation is about z-axis with an angle α.
- Second rotation is about y-axis with an angle β.

$$[P] = \cos\frac{\alpha}{2} + \sin\frac{\alpha}{2}\boldsymbol{k}, \quad [Q] = \cos\frac{\beta}{2} + \sin\frac{\beta}{2}\boldsymbol{j}$$

- As we rotate coordinate frames, the rotation operators are $L_{P^\star}$, followed by $L_{Q^\star}$, applied sequentially.
- Quaternion describing the composite rotation

$$\begin{aligned}
[PQ] &= \left(\cos\frac{\alpha}{2} + \sin\frac{\alpha}{2}\boldsymbol{k}\right)\left(\cos\frac{\beta}{2} + \sin\frac{\beta}{2}\boldsymbol{j}\right) \\
&= \cos\frac{\alpha}{2}\cos\frac{\beta}{2} + \cos\frac{\alpha}{2}\sin\frac{\beta}{2}\boldsymbol{j} + \sin\frac{\alpha}{2}\cos\frac{\beta}{2}\boldsymbol{k} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2}\boldsymbol{k}\times\boldsymbol{j} \\
&= \cos\frac{\alpha}{2}\cos\frac{\beta}{2} - \sin\frac{\alpha}{2}\sin\frac{\beta}{2}\boldsymbol{i} + \cos\frac{\alpha}{2}\sin\frac{\beta}{2}\boldsymbol{j} + \sin\frac{\alpha}{2}\cos\frac{\beta}{2}\boldsymbol{k}
\end{aligned}$$

Quaternions

- Axis of composite rotation

$$\boldsymbol{v} = \begin{bmatrix} -\sin\frac{\alpha}{2}\sin\frac{\beta}{2} \\ \cos\frac{\alpha}{2}\sin\frac{\beta}{2} \\ \sin\frac{\alpha}{2}\cos\frac{\beta}{2} \end{bmatrix}$$

- Angle of rotation

$$\cos\frac{\theta}{2} = \cos\frac{\alpha}{2}\cos\frac{\beta}{2}$$
$$\sin\frac{\theta}{2} = \|\boldsymbol{v}\|$$

- Rotational operator $L_{[PQ]^\star}$

- For unit quaternion, $\boldsymbol{p}' = [Q]^\star \boldsymbol{p}[Q]$.
- If $\boldsymbol{p} = X\boldsymbol{i} + Y\boldsymbol{j} + Z\boldsymbol{k}$ and $\boldsymbol{p}' = X'\boldsymbol{i} + Y'\boldsymbol{j} + Z'\boldsymbol{k}$ then

$$\boldsymbol{p}' = (q_0 - q_1\boldsymbol{i} - q_2\boldsymbol{j} - q_3\boldsymbol{k})\boldsymbol{p}(q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k})$$

Quaternions

$$
\begin{aligned}
\boldsymbol{p}' =& \boldsymbol{i}[X(q_0^2+q_1^2-q_2^2-q_3^2)+Y(2q_3q_0+2q_1q_2)+Z(2q_1q_3-2q_0q_2)] \\
&+\boldsymbol{j}[X(2q_1q_2-2q_3q_0)+Y(q_0^2-q_1^2+q_2^2-q_3^2)+Z(2q_1q_0+2q_3q_2)] \\
&+\boldsymbol{k}[X(2q_0q_2+2q_1q_3)+Y(2q_2q_3-2q_0q_1)+Z(q_0^2-q_1^2-q_2^2+q_3^2)]
\end{aligned}
$$

$$
\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \begin{bmatrix} q_0^2+q_1^2-q_2^2-q_3^2 & 2(q_3q_0+q_1q_2) & 2(q_1q_3-q_0q_2) \\ 2(q_1q_2-q_3q_0) & q_0^2-q_1^2+q_2^2-q_3^2 & 2(q_1q_0+q_3q_2) \\ 2(q_0q_2+q_1q_3) & 2(q_2q_3-q_0q_1) & q_0^2-q_1^2-q_2^2+q_3^2 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}
$$

$$
\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} q_0^2+q_1^2-q_2^2-q_3^2 & 2(q_1q_2-q_3q_0) & 2(q_0q_2+q_1q_3) \\ 2(q_3q_0+q_1q_2) & q_0^2-q_1^2+q_2^2-q_3^2 & 2(q_2q_3-q_0q_1) \\ 2(q_1q_3-q_0q_2) & 2(q_1q_0+q_3q_2) & q_0^2-q_1^2-q_2^2+q_3^2 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}
$$

Transformation Matrices

- Quaternion transformation matrix

$$[QT] = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_3 q_0 + q_1 q_2) & 2(q_1 q_3 - q_0 q_2) \\ 2(q_1 q_2 - q_3 q_0) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_1 q_0 + q_3 q_2) \\ 2(q_0 q_2 + q_1 q_3) & 2(q_2 q_3 - q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}$$

- Direction cosine matrix

$$[DC] = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

- Euler angle transformation matrix

$$[ET] = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \sin\phi\cos\theta \\ \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi & \cos\phi\cos\theta \end{bmatrix}$$

- Compare these three matrices and get relations among these transformations.

Reference

References

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Thank you for your attention !!!